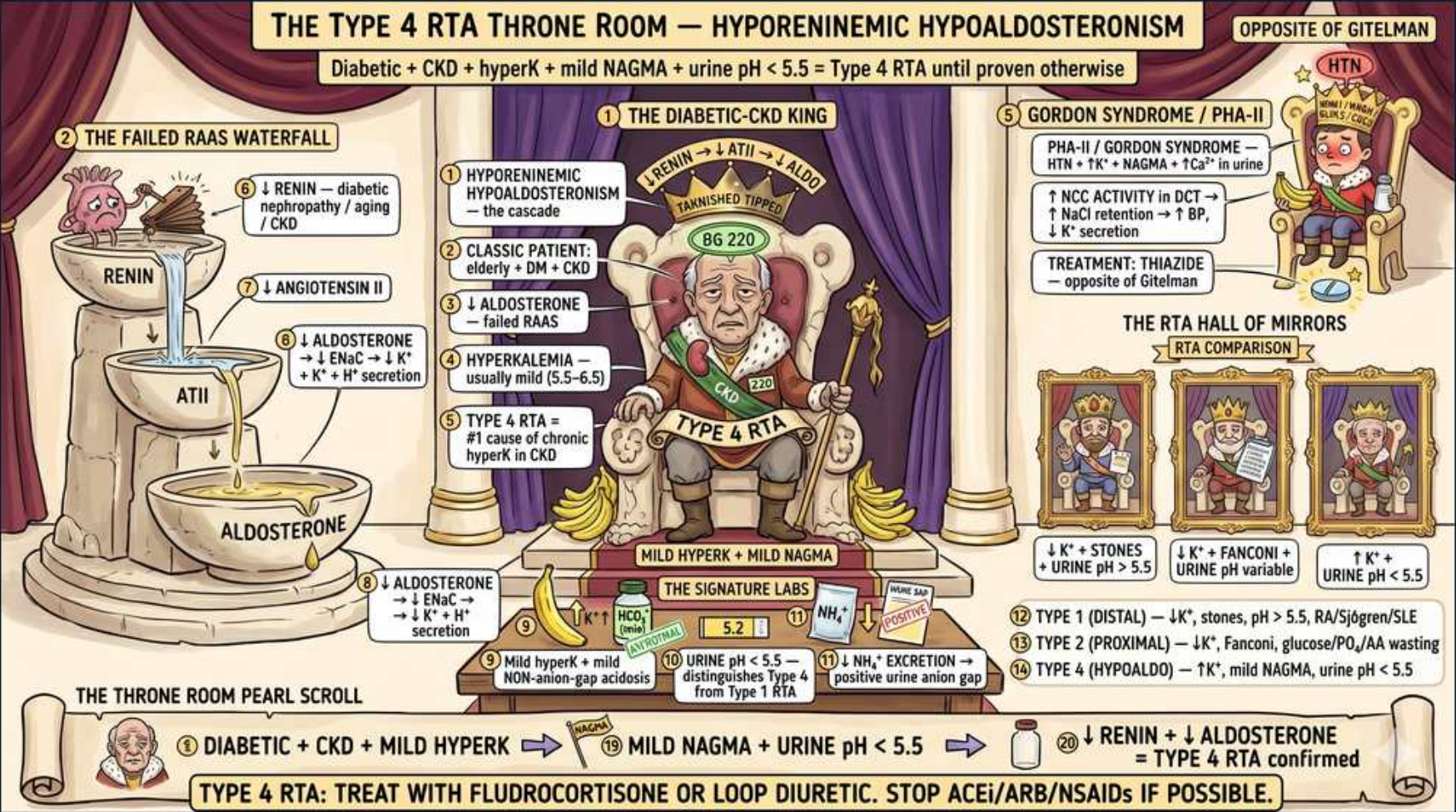


Hyperkalemia — Type 4 RTA (hyporeninemic hypoaldosteronism)



1. Type 4 = hyporenin hypoaldo

WHAT YOU SEE

Title throne: hyporeninemic hypoaldosteronism

WHAT IT ENCODES

Type 4 RTA is hyporeninemic hypoaldosteronism: low renin leads to low angiotensin II, then low aldosterone, impairing distal nephron K⁺ AND H⁺ secretion. The result is the unique combination of HYPERkalemia plus a mild non-anion-gap (hyperchloremic) metabolic acidosis. It is the most common RTA seen in adult clinical practice, especially in diabetics with CKD.

BOARD TRAP

WRONG answer: predicting hypokalemia, as in the other RTAs. Discriminator: type 4 is the ONLY renal tubular acidosis with HYPERkalemia, and it is the mechanistic opposite of Gitelman/Bartter (which waste K⁺). NAGMA + high K⁺ = type 4 until proven otherwise.

2. Diabetic + CKD

WHAT YOU SEE

The diabetic-CKD king on the throne

WHAT IT ENCODES

The prototype patient is a long-standing diabetic with chronic kidney disease who presents with persistent mild hyperkalemia and a mild metabolic acidosis (the throne shows glucose ~220). Diabetic nephropathy damages the juxtaglomerular apparatus, lowering renin release; other associations include HIV, chronic interstitial nephritis, and NSAID/ACEi/ARB use.

BOARD TRAP

WRONG answer: attributing this patient's hyperkalemia solely to 'CKD lowering GFR.' Discriminator: the K⁺ is high out of proportion to the GFR, and there is a coexisting NAGMA — that pattern is a secretory (aldosterone) defect (type 4 RTA), not simple nephron loss.

3. Low renin

WHAT YOU SEE

Failed RAAS waterfall — low renin

WHAT IT ENCODES

The cascade starts at the top: diabetic nephropathy, aging, NSAIDs, and beta-blockers reduce renin secretion from the juxtaglomerular cells (impaired conversion of prorenin to renin in diabetes). This is the 'failed waterfall' — every downstream RAAS step is starved from here.

BOARD TRAP

WRONG answer: expecting HIGH renin in this condition. Discriminator: 'hyporeninemic' is in the name — plasma renin activity is LOW. A high-renin, high-aldosterone picture points elsewhere (e.g., secondary hyperaldosteronism), and high-renin states do not cause type 4 RTA.

4. Low angiotensin II

WHAT YOU SEE

ATII level dropping

WHAT IT ENCODES

Low renin yields low angiotensin II, removing the principal stimulus for adrenal aldosterone release. ACEi and ARBs reproduce this step pharmacologically, which is exactly why those drugs precipitate or worsen type 4 RTA / hyperkalemia in susceptible patients.

BOARD TRAP

WRONG answer: thinking ACEi/ARBs are safe to continue at full dose in a hyperkalemic type 4 RTA patient. Discriminator: by further lowering angiotensin II and aldosterone, RAAS blockers aggravate the exact defect — they often must be reduced or held.

5. Low aldosterone

WHAT YOU SEE

Aldosterone bowl nearly empty

WHAT IT ENCODES

The end of the cascade: low aldosterone reduces ENaC activity and the lumen-negative voltage needed for distal K⁺ secretion (causing hyperkalemia) and reduces H⁺ secretion via the alpha-intercalated cell H⁺-ATPase (causing acidosis). Aldosterone deficiency thus produces both signature abnormalities simultaneously.

BOARD TRAP

WRONG answer: assuming aldosterone is normal because the patient isn't hypotensive or hyponatremic like Addison's. Discriminator: type 4 RTA is an ISOLATED mineralocorticoid problem with preserved cortisol/volume — patients are often euvolemic or hypertensive, unlike the volume-depleted hyponatremic Addisonian.

6. Hyperkalemia 5.5-6.5

WHAT YOU SEE

Hyperkalemia label, usually mild

WHAT IT ENCODES

Hyperkalemia in type 4 RTA is usually MILD and chronic, typically in the 5.5–6.5 mEq/L range, and is the abnormality that often unmasks the diagnosis on routine labs. It is generally well tolerated unless an acute insult (ACEi start, AKI, K⁺ load) pushes it higher.

BOARD TRAP

WRONG answer: ignoring why the acidosis is disproportionately mild given the chronicity. Discriminator: the hyperkalemia itself suppresses renal AMMONIAGENESIS, which limits NH₄⁺-based acid excretion and worsens the acidosis — correcting the K⁺ improves ammonium handling and the acidosis.

7. Most common RTA in CKD

WHAT YOU SEE

#1 cause of chronic hyperK in CKD

WHAT IT ENCODES

Type 4 RTA is the #1 cause of CHRONIC hyperkalemia in CKD, dramatically more common in clinical practice than the inherited/classic types 1 and 2. Suspect it whenever a diabetic or CKD patient has persistent mild hyperkalemia with a normal anion gap.

BOARD TRAP

WRONG answer: reaching for rare classic distal/proximal RTA diagnoses first. Discriminator: in an adult diabetic/CKD patient, type 4 is far more likely than type 1 or 2 — chase the common cause (and the drug list) before ordering an exhaustive RTA workup.

8. Mild NAGMA

WHAT YOU SEE

Signature labs: mild NAGMA

WHAT IT ENCODES

The acidosis is a mild non-anion-gap (hyperchloremic) metabolic acidosis, with serum bicarbonate usually only modestly reduced (often ~17–22 mEq/L, rarely severe). It results from impaired NH₄⁺ excretion driven by aldosterone deficiency and the suppressive effect of hyperkalemia on ammoniogenesis.

BOARD TRAP

WRONG answer: expecting a HIGH anion gap or a severe acidosis. Discriminator: a wide gap or bicarbonate in the single digits argues against uncomplicated type 4 RTA — look for an additional process (lactic acidosis, ketoacidosis, advanced uremia).

9. Urine pH < 5.5

WHAT YOU SEE

Urine pH < 5.5 on labs

WHAT IT ENCODES

In type 4 RTA the distal acidification machinery is intact, so the kidney CAN still lower urine pH below 5.5 in the face of acidosis — the problem is quantitatively too little ammonium buffer, not an inability to secrete H⁺. This preserved acidification is the key urinary discriminator.

BOARD TRAP

WRONG answer: confusing type 4 with type 1 (distal) RTA. Discriminator: type 1 distal RTA has an INABILITY to acidify (urine pH > 5.5) and HYPOkalemia, whereas type 4 acidifies normally (urine pH < 5.5) and is HYPERkalemic. The urine pH plus the K⁺ direction separates them instantly.

10. Low NH₄⁺ excretion

WHAT YOU SEE

Low urine NH₄⁺, positive urine anion gap

WHAT IT ENCODES

Aldosterone deficiency and hyperkalemia impair renal ammoniogenesis and NH₄⁺ excretion. Because urine NH₄⁺ is hard to measure directly, a POSITIVE urine anion gap (Na⁺ + K⁺ – Cl[–]) serves as a surrogate for low NH₄⁺ — confirming the renal (RTA) origin of a hyperchloremic acidosis rather than GI bicarbonate loss.

BOARD TRAP

WRONG answer: using a positive urine anion gap to diagnose diarrhea. Discriminator: a NEGATIVE urine anion gap (high NH₄⁺) indicates an appropriate renal response, as in diarrhea; a POSITIVE urine anion gap (low NH₄⁺) indicates an RTA. Type 4 gives a positive urine anion gap.

11. Gordon / PHA-II

WHAT YOU SEE

Gordon syndrome panel, treat with thiazide

WHAT IT ENCODES

Gordon syndrome (pseudohypoaldosteronism type II, familial hyperkalemic hypertension) is a WNK-kinase defect that over-activates the thiazide-sensitive NCC cotransporter, causing increased NaCl reabsorption, volume expansion, HYPERTENSION, hyperkalemia, and a mild NAGMA — with LOW renin but normal/high aldosterone. The elegant treatment is a thiazide diuretic, which blocks the overactive transporter.

BOARD TRAP

WRONG answer: confusing Gordon (PHA-II) with PHA-I or with type 4 RTA. Discriminator: Gordon has HYPERTENSION + high/normal aldosterone and responds to thiazides; type 4 RTA is hypoaldosteronism. The mirror image is Gitelman syndrome (loss-of-function NCC) — low BP, hypokalemia, metabolic alkalosis.

12. RTA comparison

WHAT YOU SEE

RTA hall of mirrors: types 1/2/4

WHAT IT ENCODES

Side-by-side: Type 1 (distal) — HYPOkalemia, inability to acidify (urine pH > 5.5), calcium phosphate stones/nephrocalcinosis, severe acidosis, treat with bicarbonate. Type 2 (proximal) — HYPOkalemia, bicarbonate wasting with Fanconi syndrome (glucosuria, phosphaturia, aminoaciduria), urine pH variable. Type 4 — HYPERkalemia, mild NAGMA, urine pH < 5.5, hypoaldosteronism.

BOARD TRAP

WRONG answer: mixing up which RTAs waste vs retain potassium. Discriminator: types 1 and 2 are HYPOkalemic; only type 4 is HYPERkalemic. The single fastest exam screen: NAGMA + low K⁺ = type 1 or 2; NAGMA + high K⁺ = type 4.

13. Stop ACEi/ARB/NSAID

WHAT YOU SEE

Pearl scroll: stop offending drugs

WHAT IT ENCODES

First-line management is to remove or reduce drugs that worsen hypoaldosteronism/hyperkalemia: ACEi, ARBs, NSAIDs, potassium-sparing diuretics (spironolactone/eplerenone/amiloride), trimethoprim, and heparin. Add dietary potassium restriction and avoid salt substitutes (which are KCl).

BOARD TRAP

WRONG answer: continuing an NSAID or ACEi unchanged while treating the K⁺ symptomatically. Discriminator: failing to address the offending drugs leaves the underlying RAAS suppression in place — drug review is the highest-yield, lowest-cost intervention before adding new agents.

14. Fludrocortisone / loop

WHAT YOU SEE

Bottom banner treatment

WHAT IT ENCODES

Definitive therapy replaces or bypasses the missing mineralocorticoid effect: fludrocortisone (a synthetic mineralocorticoid, e.g., 0.1 mg daily) restores ENaC-driven K⁺ and H⁺ secretion, while a loop or thiazide diuretic lowers K⁺ and helps correct the acidosis (useful when volume overload/HTN makes fludrocortisone risky). Sodium bicarbonate can be added for persistent acidosis.

BOARD TRAP

WRONG answer: giving fludrocortisone to a hypertensive, volume-overloaded type 4 RTA patient by reflex. Discriminator: fludrocortisone causes sodium/water retention and can worsen HTN and edema — in those patients favor a loop diuretic (and/or bicarbonate) over mineralocorticoid replacement.